knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE) library(tidyverse) library(janitor)

—- Read CSV files —-

rep_2018 <- read_csv(“00_rep_candidates_2018.csv”) rep_2022 <- read_csv(“00_rep_candidates_2022.csv”) dem_2018 <- read_csv(“00_dem_candidates_2018.csv”) dem_2022 <- read_csv(“00_dem_candidates_2022.csv”)

—- Clean column names —-

rep_2018 <- rep_2018 %>% clean_names() rep_2022 <- rep_2022 %>% clean_names() dem_2018 <- dem_2018 %>% clean_names() dem_2022 <- dem_2022 %>% clean_names()

—- Convert primary_percent to numeric —-

rep_2018 <- rep_2018 %>% mutate(primary_percent = as.numeric(gsub(“[ˆ0-9.]”, “ “, primary_percent))) rep_2022 <- rep_2022 %>% mutate(primary_percent = as.numeric(gsub(”[ˆ0-9.]“,”“, primary_percent))) dem_2018 <- dem_2018 %>% mutate(primary_percent = as.numeric(gsub(”[ˆ0-9.]“,”“, primary_percent))) dem_2022 <- dem_2022 %>% mutate(primary_percent = as.numeric(gsub(”[ˆ0-9.]“,” “, primary_percent)))

—- Drop empty rows/columns —-

rep_2018 <- rep_2018 %>% remove_empty(c(“rows”, “cols”)) %>% drop_na(candidate, state) rep_2022 <- rep_2022 %>% remove_empty(c(“rows”, “cols”)) %>% drop_na(candidate, state) dem_2018 <- dem_2018 %>% remove_empty(c(“rows”, “cols”)) %>% drop_na(candidate, state) dem_2022 <- dem_2022 %>% remove_empty(c(“rows”, “cols”)) %>% drop_na(candidate, state)

—- Combine datasets —-

)“‘ title: How Demographics and Party Dynamics Shape Primaries —

Abstract: This project has the goal of exploring the relationship between party dynamics, candidate demographics, and the polarization of ideology in the US. Our group is going to utilize the datasets of the Democratic and Republican primary candidates from 2018/2022 election cycles in addition to congressional polarization measures to answer key questions. With the data of the preliminary results, we can see that while the two parties have definitely made progress in diversifying the candidate pools, there’s still variance in success rates grouped by gender, race, and office type – there’s also strong influence being placed by party affiliation and incumbency. It is also shown that significant drops in ideological overlap in Congress correlates with significant shifts in primary competitiveness; we will explore this further and our future analyses will utilize modeling to better quantify these relationships between demographic and primary outcomes.

Intro:

The purpose of this project is to examine how gender, race, and ideology correlate with political success in U.S. primary elections and how they differ among factors like parties. Primary elections are important to utilize when determining who should represent a certain party in the election. We will be focusing on two questions. Our first question is: “How do gender and race correlate with primary success rates, and do these patterns differ by party or office (Senate, House, Governor)?” Next, we’re going to ask: “How does the ideological overlap between Democrats and Republicans in Congress change around periods of large shifts in primary dynamics like ‘wave years’?” To answer these questions, we will utilize logistic regression, which is utilized to predict an outcome's likelihood for events such as winning a primary. To see which traits are significant for each party's candidates, we can compare separate models made for Republicans and Democrats.

Data:

We are using five datasets: 4 of them revolve around candidate data (Democrats and Republicans) from 2018 and 2022 with information like name, race, gender, incumbency status and if they won their primary and the last dataset is on polarization which will show how much ideological overlap there is between Democrats and Republicans over time. We cleaned the data by removing empty/null columns and also changed the names of columns to string only, to make referring to them more efficient. We also combined the datasets into one dataframe "all_cands" to be better at pulling data.

Visualizations:

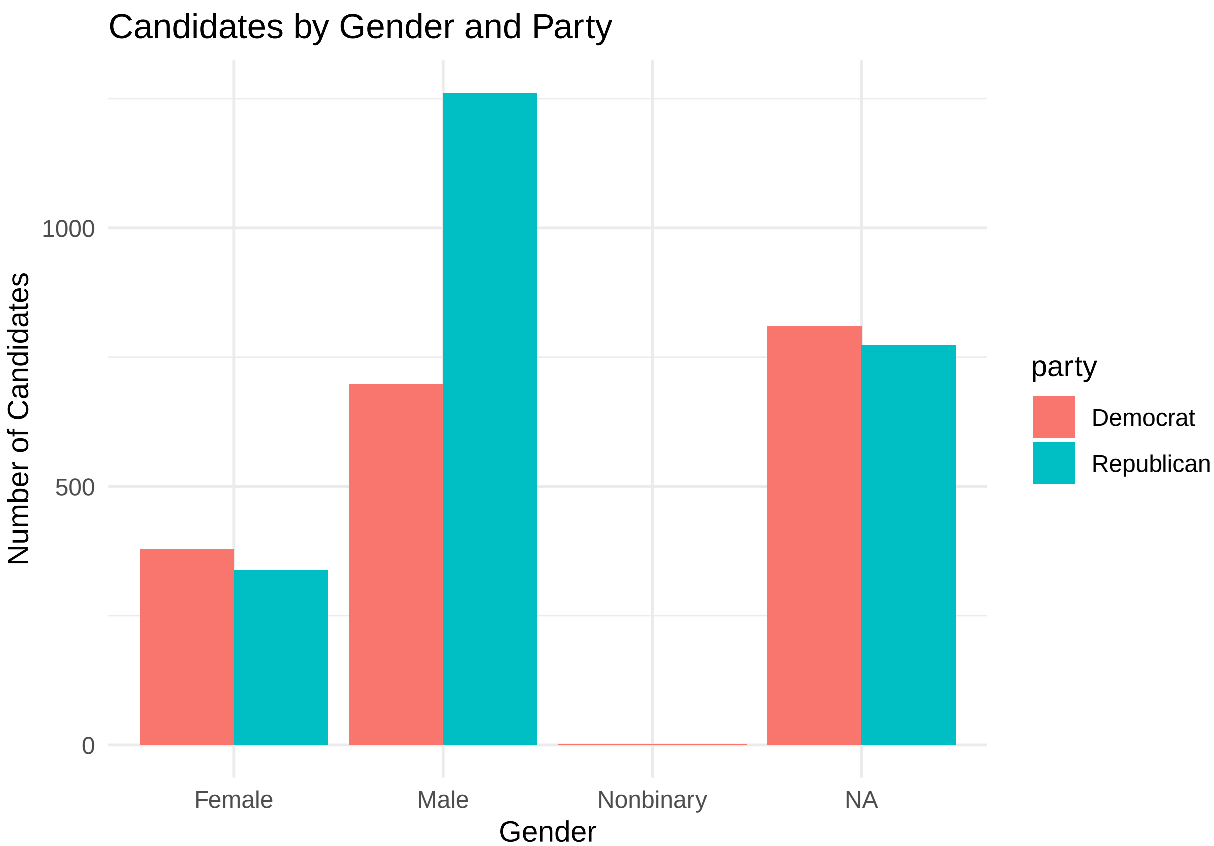
candidates <- read_csv("data/cleaned_candidates.csv") polarization <- read_csv("data/00_Political Polarization in US Congress (Assorted).csv")

```r
# --- Load libraries ---
library(tidyverse)



# --- Optional: print summary ---
glimpse(candidates)
```

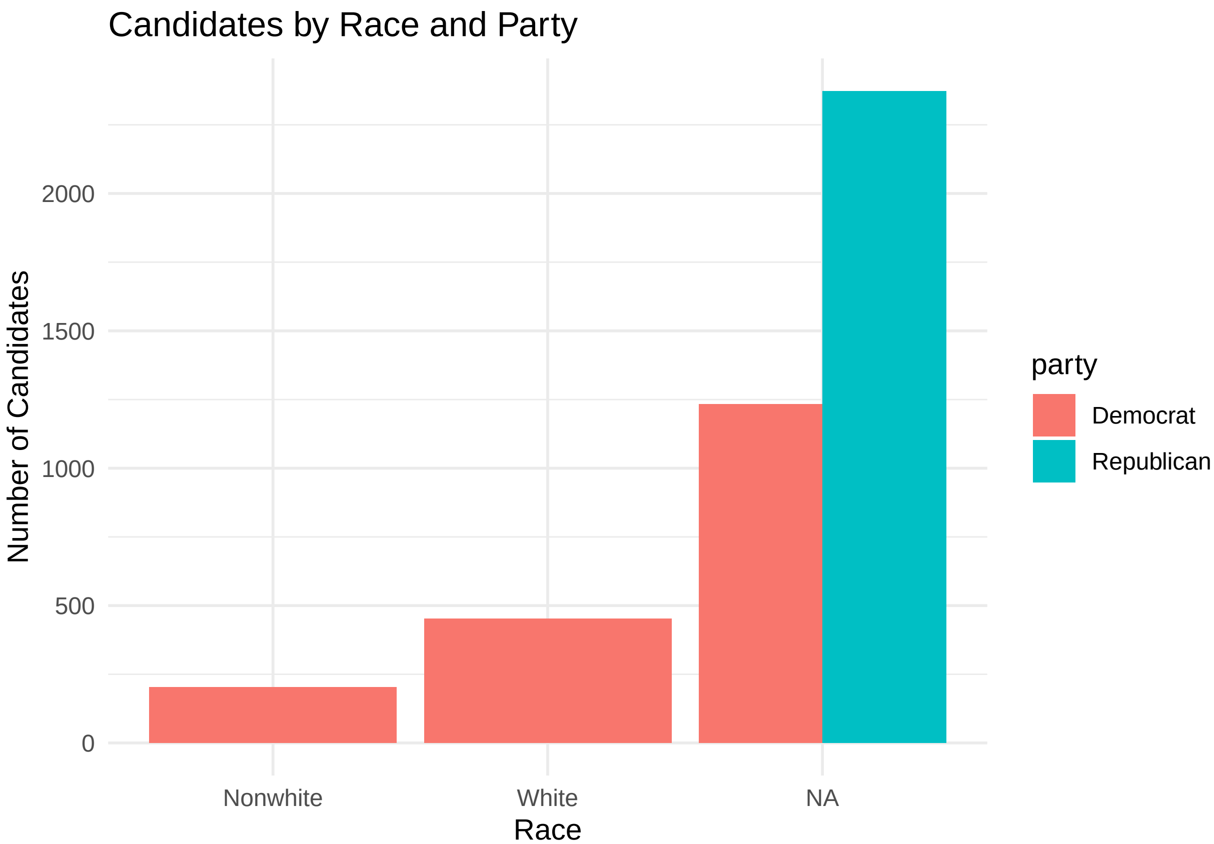
```
## Rows: 4,261
## Columns: 78
## $ candidate                   <chr> "Mike Dunleavy", "Michael Sheldon", "Mead T~
## $ state                       <chr> "AK", "AK", "AK", "AK", "AK", "AK", "AK", "~
## $ district                    <chr> "Governor of Alaska", "Governor of Alaska",~
## $ office_type                 <chr> "Governor", "Governor", "Governor", "Govern~
## $ race_type                   <chr> "Regular", "Regular", "Regular", "Regular",~
## $ race_primary_election_date <chr> "8/21/18", "8/21/18", "8/21/18", "8/21/18",~
## $ primary_status              <chr> "Advanced", "Lost", "Lost", "Lost", "Lost",~
## $ primary_runoff_status       <chr> "None", "None", "None", "None", "None", "No~
## $ general_status              <chr> "On the Ballot", "None", "None", "None", "N~
## $ primary_percent             <dbl> 61.80, 2.20, 31.90, 0.60, 1.30, 1.50, 0.70,~
## $ won_primary                 <chr> "Yes", "No", "No", "No", "No", "No", "No", ~
## $ rep_party_support           <chr> NA, NA, NA, NA, NA, NA, NA, "Yes", "Yes", "~
## $ trump_endorsed              <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ bannon_endorsed             <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ great_america_endorsed      <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ nra_endorsed                <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ right_to_life_endorsed      <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ susan_b_anthony_endorsed    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ club_for_growth_endorsed    <chr> NA, NA, NA, NA, NA, NA, NA, "Yes", "No", "N~
## $ koch_support                <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, "Ye~
## $ house_freedom_support       <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ tea_party_endorsed          <chr> NA, NA, NA, NA, NA, NA, NA, "Yes", "No", "N~
## $ main_street_endorsed        <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ chamber_endorsed            <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ no_labels_support           <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ party                       <chr> "Republican", "Republican", "Republican", "~
```

```
## $ year                    <dbl> 2018, 2018, 2018, 2018, 2018, 2018, 2018, 2~
## $ gender                  <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ race_1                  <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ race_2                  <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ race_3                  <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ incumbent               <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ incumbent_challenger    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ primary_date            <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ office                  <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ primary_votes           <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ primary_outcome         <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ runoff_votes            <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ runoff_percent          <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ runoff_outcome          <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ x2020_election_stance   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ trump                   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ trump_date              <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ club_for_growth         <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ party_committee         <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ renew_america           <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ e_pac                   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ view_pac                <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ maggies_list            <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ winning_for_women       <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ partisan_lean           <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ race                    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ veteran                 <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ lgbtq                   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ elected_official        <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ self_funder             <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ stem                    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ obama_alum              <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ party_support           <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ emily_endorsed          <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ guns_sense_candidate    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ biden_endorsed          <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ warren_endorsed         <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ sanders_endorsed        <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ our_revolution_endorsed <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ justice_dems_endorsed   <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ pccc_endorsed           <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ indivisible_endorsed    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ wfp_endorsed            <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ vote_vets_endorsed      <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ emil_ys_list            <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ justice_dems            <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ indivisible             <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ pccc                    <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ our_revolution          <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ sunrise                 <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ sanders                 <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ aoc                     <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
```

```r
# --- Histogram / Bar charts: Candidates by gender and race ---
ggplot(candidates, aes(x = gender, fill = party)) +
  geom_bar(position = "dodge") +
  labs(title = "Candidates by Gender and Party",
       x = "Gender", y = "Number of Candidates") +
  theme_minimal()
```




```r
ggplot(candidates, aes(x = race, fill = party)) +
  geom_bar(position = "dodge") +
  labs(title = "Candidates by Race and Party",
       x = "Race", y = "Number of Candidates") +
  theme_minimal()
```




```
# --- Boxplot: Ideology by primary success ---
##ggplot(candidates, aes(x = factor(won_primary, labels = c("Lost", "Won")),
   ##                    y = ideology_score, fill = party)) +
 ## geom_boxplot() +
  ##labs(title = "Ideology by Primary Outcome",
  ##     x = "Primary Outcome", y = "Ideology Score") +
 ## theme_minimal()
```

The histograms relate to primary dynamics, which is how competitive and dynamic the primaries are. The histograms tell us that the primaries were actyally not as competitive, as there's typically one strong, unchallenged candidate. The wave elections would show a lot more spread in the middle range – this is the candidates splitting the votes and it seems very spread out. The difference between the 2018 and 2022 primaries show us how polarized they were individually. The x-axis represents the candidate's share of votes in their primary ranged from0 to 100% and the y-axis represents how many candidates got that share. There are visibly two spikes – one spike is near 0% which means many Democrats got extremely low support and lost badly and another spike is near 100%, which means there were candidates who were strong and won overwhelmingly. So, we can conclude that most democratic primaries had either very strong, practically unopposed candidates or hopeless ones with small to no votes. Very few of the races were neck and neck.

For the line plot, the x-axis represents the election year and the y-axis represents the average share of votes that each cadidate got in their party's primaries. Looking at the trend of the graph, there is a very strong rise from 31% to 36%; this suggests candidates received a larger portion of votes in 2022 than 2018 representing more dominant winners (incumbents) than close races. This was also supported by the histograms. We can conclude from this that primaries grew to be less competitive in 2022 than they were in 2018.

Analysis:

We chose to use logistic regression to take a deeper look at how different factors can be used to predict success in a primary election. It's especially helpful because it works with binary outcomes (win/lose) and it lets us know how much the odds of winning or losing gets affected by a particular variable. To do so, we built one model that includes all candidates:

P(Win)= 0 + 1(Party) + 2(Gender) + 3(Race) + 4(Ideology) + 5(Incumbent) +

P(Win) represents the probability that a candidate will win their primary, B0 is the intercept which in this case is the baseline odds a candidate has of winning, B1 to B5 are the coefficients for each variable and shows how much each factor will increase/decrease odds of winning. This model will help us predict success through other variables besides party. We utilized another model to separate democrats and republicans:

P(Win) = 0 + 1(Gender) + 2(Race) + 3(Ideology) + 4(Incumbent) +

This model will help us compare results to see what factors are more influential to a certain party. For example, if in the democratic model, gender has a stronger effect, it may mean that Democratic voters are more likely to support female candidates. We will also check how well the models fit the data through R^2 and visual checks.

Future work We'll include testing to see whether certain effects end up changing due to their party – an example would be if being a woman is more beneficial among Democrats vs Republicans. We will also try using more models like decision trees to see if our results are confirmed and if these trends stay consistent.

Author Contribution:

Ananya: wrote the report, worked on visualizations

Melody: worked on the logistic regression, formatting

Chiko: worked on cleaning data, regression, visualizations